Alex Liu

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Education

Georgia Institute of Technology, *M.S. in Computer Science* ***01/2025 - Present***

Carnegie Mellon University, *M.S. in Information Systems* ***01/2021 - 12/2022***

University of Colorado, *B.S. in Economics and Minor in Mathematics* ***09/2016 - 05/2020***

Experience

Research Assistant, Georgia Institute of Technology ***05/2026 - Present***

- Profiled **PyTorch/GPU** pipelines and wrote fused **CUDA** kernels for a Transformer-based vision-language-action (VLA) policy, cutting onboard inference latency 1.5× on edge hardware
- Built a real-time **C++** safety layer on Linux bounding model actions by workspace and velocity limits, with shared-memory IPC and a deterministic fail-safe stop path
- Built a **Python** simulation harness to test fault cases like sensor dropout and stale messages, tracking success and safety-violation rates as weekly regression metrics

Software Engineer, Pennsylvania Turnpike Commission ***06/2023 - 12/2024***

- Engineered real-time **C++/Linux** roadside systems across 500+ highway miles, fusing heterogeneous sensor inputs and processing 500K+ daily transactions
- Built **Python/SQ**L analysis and dashboards tracking failure and exception rates fleet-wide across 2M+ monthly transactions, isolating root causes behind 13% revenue leakage
- Designed an automated escalation path routing low-confidence perception outputs to human operator review queues via Azure Service Bus, improving approval rate by 12%
- Shipped an end-to-end license plate recognition pipeline on Linux edge nodes, with a **Python/FastAPI** service ingesting 67K+/day camera images in Dockerized microservices
- Profiled and optimized the DMV lookup pipeline from 400ms to ~50ms via Elasticsearch fuzzy matching and Redis caching, improving plate accuracy by 3%

Software Engineer Intern, SAP ***07/2020 - 12/2020***

- Expanded automated test coverage with JUnit and Mockito unit and integration suites, raising line coverage from 50% to 80% across 22 Onboarding components
- Validated migrated OData endpoints in Java against legacy API behavior, verifying response schema and data parity across job requisition flows before cutover

Skills

Languages: Java, Python, Go, C++, JavaScript, TypeScript, HTML, CSS

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis, Elasticsearch

Cloud & DevOps: Azure, AWS (ECS, RDS, S3), Docker, Kubernetes, Terraform

AI/ML: LLM API Integration, ML Infrastructure, Computer Vision, LangChain, MCP